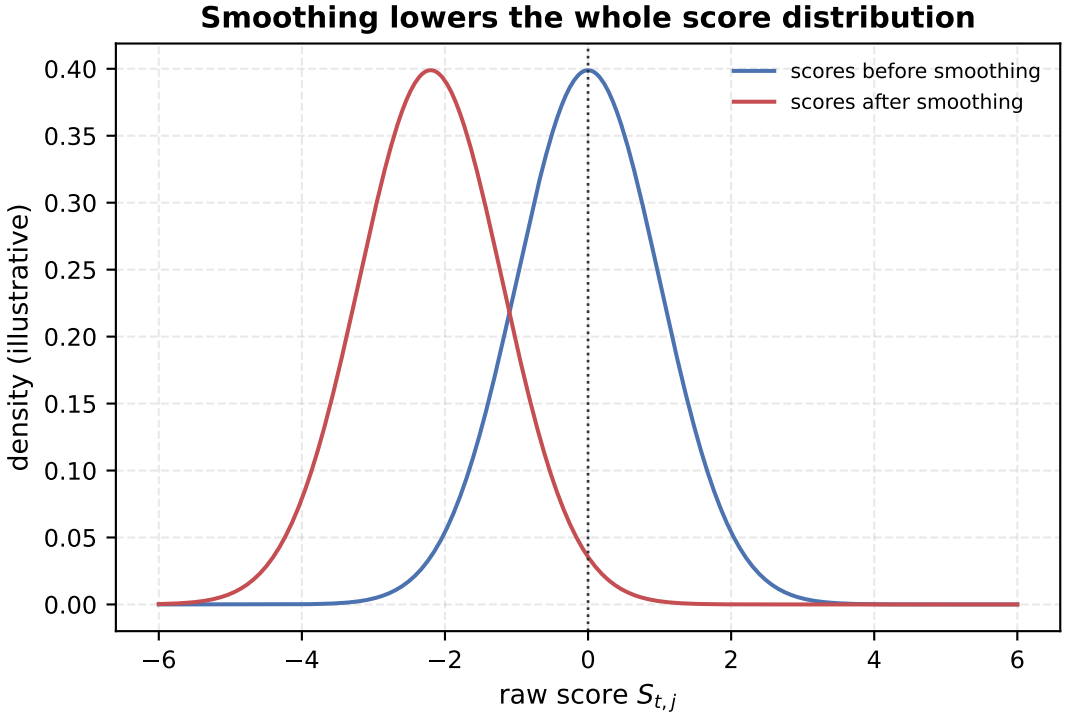

Why the shift is invisible to Softmax but fatal to rectification

Softmax	$S \to S + c$ leaves W unchanged *translation-invariant: the shift cancels*
Softmax + FIR	W unchanged by the shift *but the mean score is still lower*
Power-norm + FIR	$(S')_+^p$ is not translation-invariant *the shift removes mass below the threshold*
Power-norm + FIR + tau	τ is learned per layer *the threshold restores the mass*